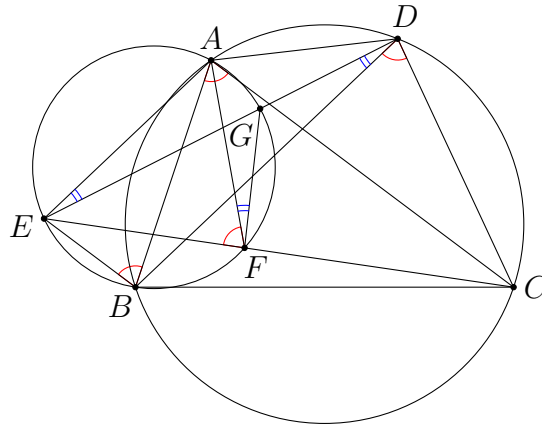


Circles — Solutions

Problem 1. (MEMO 2015 Individual Problem 3) Let $ABCD$ be a cyclic quadrilateral. Let E be the intersection of lines parallel to AC and BD passing through points B and A , respectively. The lines EC and ED intersect the circumcircle of triangle AEB again at F and G , respectively. Prove that points C, D, F , and G lie on a circle.



Solution. We have

$$\angle AFE = \angle ABE = \angle BAC = \angle BDC$$

and

$$\angle GFA = \angle GEA = \angle EDB.$$

Adding these, we obtain $\angle GFE = \angle AFE + \angle GFA = \angle BDC + \angle EDB = \angle EDC$. Thus, C, D, G, F are concyclic. $\square$

Problem 2. (IberoAmerican MO 2019 Problem 4) Let $ABCD$ be a trapezoid with $AB \parallel CD$ inscribed in a circle Γ . Let P and Q be two points on segment AB (where A, P, Q, B appear in that order and are distinct) such that $AP = QB$. Let E and F be the second intersection points of lines CP and CQ with Γ , respectively. Lines AB and EF intersect at G . Prove that line DG is tangent to Γ .

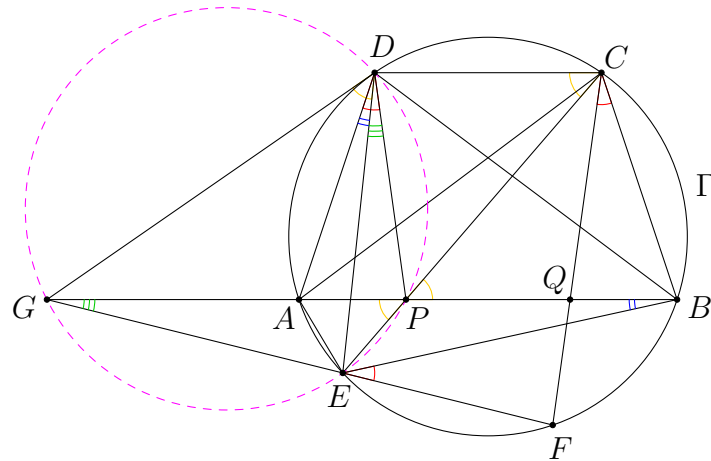
Solution. Note that $ABCD$ is an isosceles trapezoid since it is cyclic. It is then obvious that $\triangle DAP \cong \triangle CBQ$. Therefore, we have $\angle ADP = \angle QCB = \angle FEB$. It follows that

$$\angle EDP = \angle ADP - \angle ADE = \angle FEB - \angle ABE = \angle EGP.$$

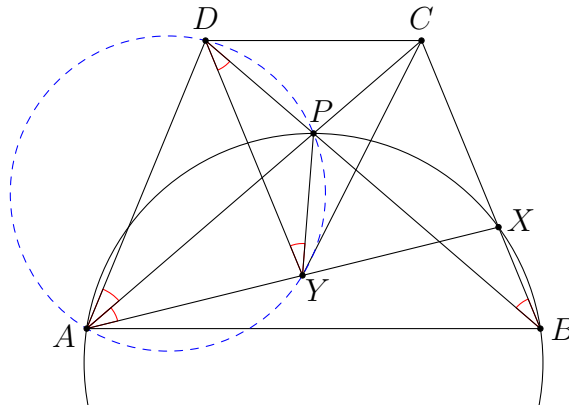
Therefore, D, G, E, P are concyclic. Thus, we have

$$\angle GDE = \angle GPE = \angle BPC = \angle DCE.$$

This implies GD is tangent to Γ as desired. $\square$



Problem 3. (Iran MO 2nd Round 2018 Problem 1) Let P be the intersection of AC and BD in isosceles trapezoid $ABCD$ (with $AB \parallel CD$ and $BC = AD$). The circumcircle of triangle ABP intersects BC for the second time at X . Point Y lies on AX such that $DY \parallel BC$. Prove that $\angle YDA = 2\angle YCA$.

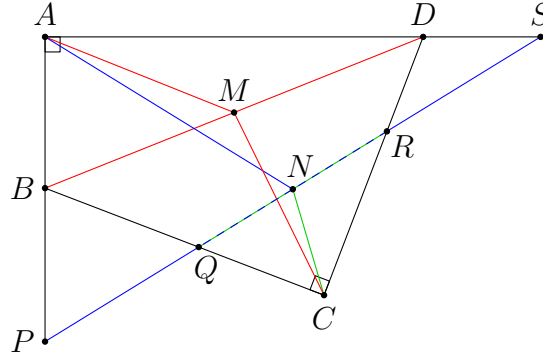


Solution. Firstly, from $\angle YDP = \angle XBP = \angle YAP$, the points D, A, Y, P are concyclic. It follows that

$$\angle PDY = \angle DBC = \angle CAD = \angle PYD.$$

Therefore, $PY = PD$, and hence $PY = PC$. This implies $\angle ADY = \angle APY = 2\angle ACY$. $\square$

Problem 4. (EGMO 2017 Problem 1) Let $ABCD$ be a convex quadrilateral with $\angle DAB = \angle BCD = 90^\circ$ and $\angle ABC > \angle CDA$. Let Q and R be points on segments BC and CD , respectively, such that line QR intersects lines AB and AD at points P and S , respectively. It is given that $PQ = RS$. Let the midpoint of BD be M and the midpoint of QR be N . Prove that the points M, N, A and C lie on a circle.

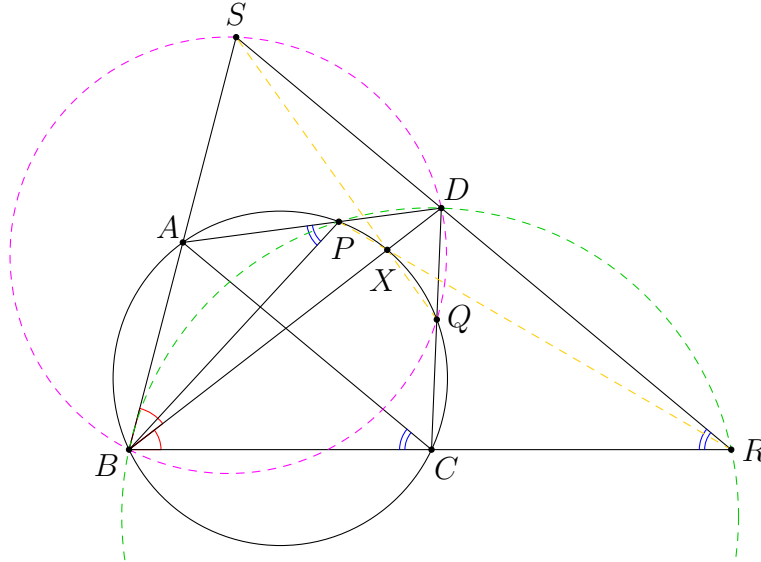


Solution. Firstly, note that $MA = MB = MC = MD$ as a property of right-angled triangles. So M is the centre of the circle passing through A, B, C, D . Similarly, we have $NC = NQ = NR$. Also, as $NP = NQ + QP = NR + RS = NS$, we have $NA = NP = NS$. Using these isosceles triangles, we obtain

$$\angle ANC = \angle ANP + \angle QNC = 2\angle ASP + 2\angle QRC = 2\angle DSR + 2\angle SRD = 2\angle ADC = \angle AMC.$$

This implies A, M, N, C are concyclic. $\square$

Problem 5. (Baltic Way 2014 Problem 14) Let $ABCD$ be a convex quadrilateral such that the line BD bisects the angle ABC . The circumcircle of triangle ABC intersects the sides AD and CD in the points P and Q , respectively. The line through D and parallel to AC intersects the lines BC and BA at the points R and S , respectively. Prove that the points P, Q, R and S lie on a common circle.

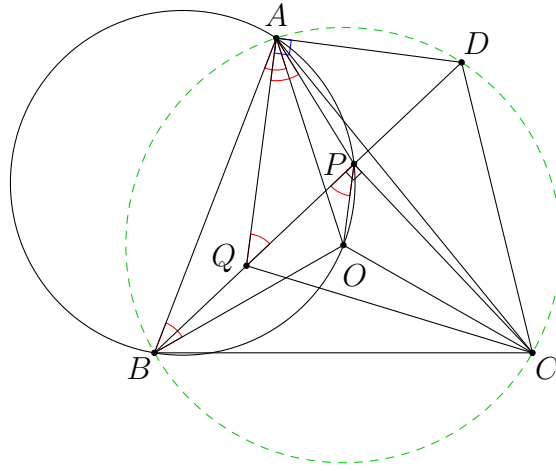


Solution. From $\angle APB = \angle ACB = \angle DRB$, the points B, R, D, P are concyclic. Similarly, B, S, D, Q are concyclic. Let BD meet PR at X . As

$$\angle BXP = \angle RBX + \angle XRB = \angle DBA + \angle PDB = \angle PAS,$$

the points A, B, X, P are concyclic. Therefore, X is the second intersection of BD and (ABC) . By symmetry, QS also passes through X . Thus, we have $XP \times XR = XB \times XD = XQ \times XS$, which implies P, Q, R, S are concyclic. $\square$

Problem 6. (MEMO 2016 Individual Problem 3) Let ABC be an acute triangle such that $\angle BAC > 45^\circ$ with circumcentre O . A point P is chosen inside triangle ABC such that A, P, O, B are concyclic and the line BP is perpendicular to the line CP . A point Q lies on the segment BP such that the line AQ is parallel to the line PO . Prove that $\angle QCB = \angle PCO$.



Solution. Note that

$$\angle PAQ = 180^\circ - \angle APO = \angle ABO = \angle BAO = \angle BPO = \angle PQA.$$

Let D be the point such that P is the midpoint of DQ . Since $PD = PQ = PA$, we have $AD \perp AQ$. This implies $AD \perp OP$. As $PA = PD$, the line OP is the perpendicular bisector of AD . Therefore, D lies on the circumcircle of $\triangle ABC$.

Next, we have $\triangle CPQ \cong \triangle CPD$ since $CP = CP$, $PQ = PD$ and $\angle CPQ = \angle CPD = 90^\circ$. It follows that

$$\angle CQP = \angle CDP = \angle CDB = \angle CAB,$$

and hence $\angle PCQ = 90^\circ - \angle CQP = 90^\circ - A = \angle OCB$. Thus,

$$\angle QCB = \angle OCB - \angle OCQ = \angle PCQ - \angle OCQ = \angle PCO.$$

$\square$

Problem 7. (Iran MO 2nd Round 2016 Problem 2) Let ABC be a triangle such that $\angle C = 2\angle B$ and ω be its circumcircle. A tangent from A to ω intersects BC at E . Ω is a circle passing through B that is tangent to AC at C . Let Ω intersect AB again at F . K is a point on Ω such that EK is tangent to Ω (where A and K are on different sides of BC). Let M be the midpoint of arc BC of ω (not containing A). Prove that $AFMK$ is a cyclic quadrilateral.



by the angle bisector theorem. We shall show that F, D, K are collinear. Since $\angle FBC = \angle FCB$, FK is the internal angle bisector of $\angle BKC$. Therefore, it suffices to show DK bisects $\angle BKC$. Equivalently, we have to show $\frac{KB}{KC} = \frac{DB}{DC}$, or

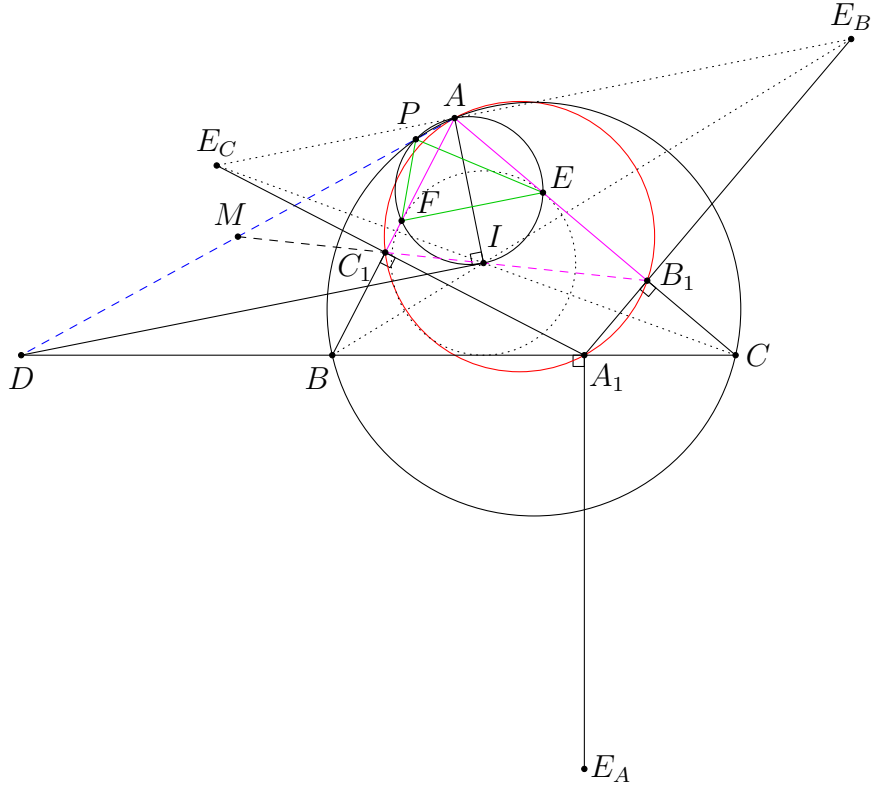
Now, from $\triangle EBK \sim \triangle EKC$, we have $\frac{EB}{EK} = \frac{EK}{EC} = \frac{BK}{KC}$. This implies

Similarly, from $\triangle EBA \sim \triangle EAC$, we obtain

This proves our claim that F, D, K are collinear. Finally, we have

and hence A, F, M, K are concyclic.

Problem 8. (USA TST 2019 Problem 6) Let ABC be a triangle with incentre I , and let D be a point on line BC satisfying $\angle AID = 90^\circ$. Let the excircle of triangle ABC opposite the vertex A be tangent to BC at A_1 . Define points B_1 on CA and C_1 on AB analogously, using the excircles opposite B and C , respectively. Prove that if quadrilateral $AB_1A_1C_1$ is cyclic, then AD is tangent to the circumcircle of $\triangle DB_1C_1$.



Solution. Let E_A, E_B, E_C be the A, B, C -excentres of $\triangle ABC$ respectively. It is well-known that $E_A A_1, E_B B_1, E_C C_1$ are concurrent at some point X . We claim that $X = A_1$. Suppose not, then X lies on $(AC_1 A_1 B_1)$ since $\angle AC_1 X = \angle AB_1 X = 90^\circ$. Also, this shows AX is a diameter of the circle, and hence $\angle AA_1 X = 90^\circ$. But since $A_1 X \perp BC$, it forces $AA_1 \parallel BC$, contradiction. Therefore, we must have $X = A_1$. Applying Pappus' theorem to the lines $E_C A E_B$ and $BA_1 C$, we know that $E_C A_1 \cap AB = C_1, E_C C \cap E_B B = I$ and $AC \cap E_B A_1 = B_1$ are collinear.

Next, let E and F be the contact points of the incircle with the sides AC and AB respectively. Let P be the second intersection point of (ABC) and (AEF) . Thus, P is the centre of spiral similarity mapping BF to CE . This implies

$$\frac{PF}{PE} = \frac{BF}{CE} = \frac{AC_1}{AB_1}.$$

Together with $\angle FPE = \angle FAE = \angle C_1 A B_1$, we have $\triangle PFE \sim \triangle AC_1 B_1$.

From $\angle DIB = \angle AIB - 90^\circ = \angle ICB$, we see that DI is tangent to (BIC) . Note that DI is also tangent to (AEF) since the centre of (AEF) is the midpoint of AI . Thus, D is the radical centre of (AEF) , (BIC) and (ABC) , meaning A, P, D are collinear. Therefore, we obtain

$$\angle DAC_1 = \angle PAF = \angle PEF = \angle AB_1 C_1.$$

This shows AD is tangent to $(AB_1 C_1)$.

Now, let M be the midpoint of AD . If we can show that M lies on $B_1 C_1$, then from

$$MD^2 = MA^2 = MC_1 \times MB_1,$$

we can deduce AD is tangent to (DB_1C_1) . To prove this, note that $MA = MI$ as $\angle AID = 90^\circ$. Therefore, we find that

$$\angle AIM = \angle MAI = \angle PAF + \angle FAI = \angle PEF + \frac{A}{2} = \angle AB_1C_1 + \angle IAB_1 = \angle AIC_1.$$

This implies M, C_1, I are collinear and so the proof is complete. □